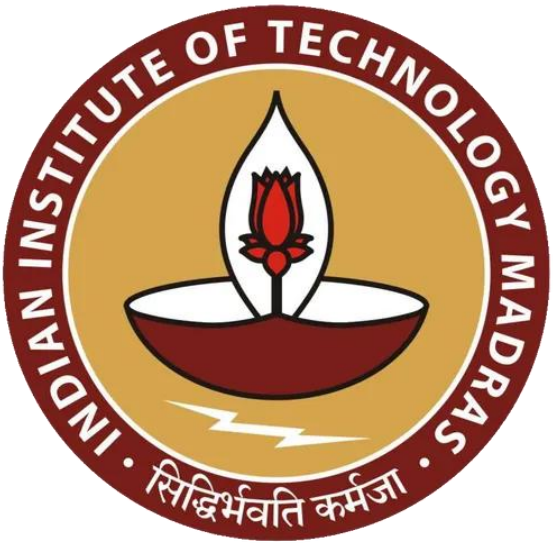
# Data-Driven Optimization for Inventory, Credit Risk, and Supply Chain Management in New-A-One & Co.

*Mid-Term Submission for the BDM Capstone Project*

Submitted by
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# 1) Executive Summary

This capstone project addresses critical operational challenges at New-A-One & Co., a wholesale distributor in Firozabad, India, through data-driven optimization of inventory, credit risk, and supply chain management. The business faces three key issues: (1) inventory mismanagement, with overstocking of slow-moving designs and stockouts of high-demand items ; (2) credit risks from delayed payments and (3) supply chain inefficiencies, including inconsistent supplier delivery times and quality.

Using six months of operational data (June–December 2024), the project implements analytical solutions to enhance decision-making. Inventory analysis reveals that AL designs account for 44% of sales but face frequent stockouts, while AH designs contribute to 15% wastage. Credit risk assessment identifies ₹2.27 lakhs in pending payments, with Owais Bangle Store having 100% unpaid dues. Supplier evaluation highlights NavJivan Glass Works as the most reliable vendor

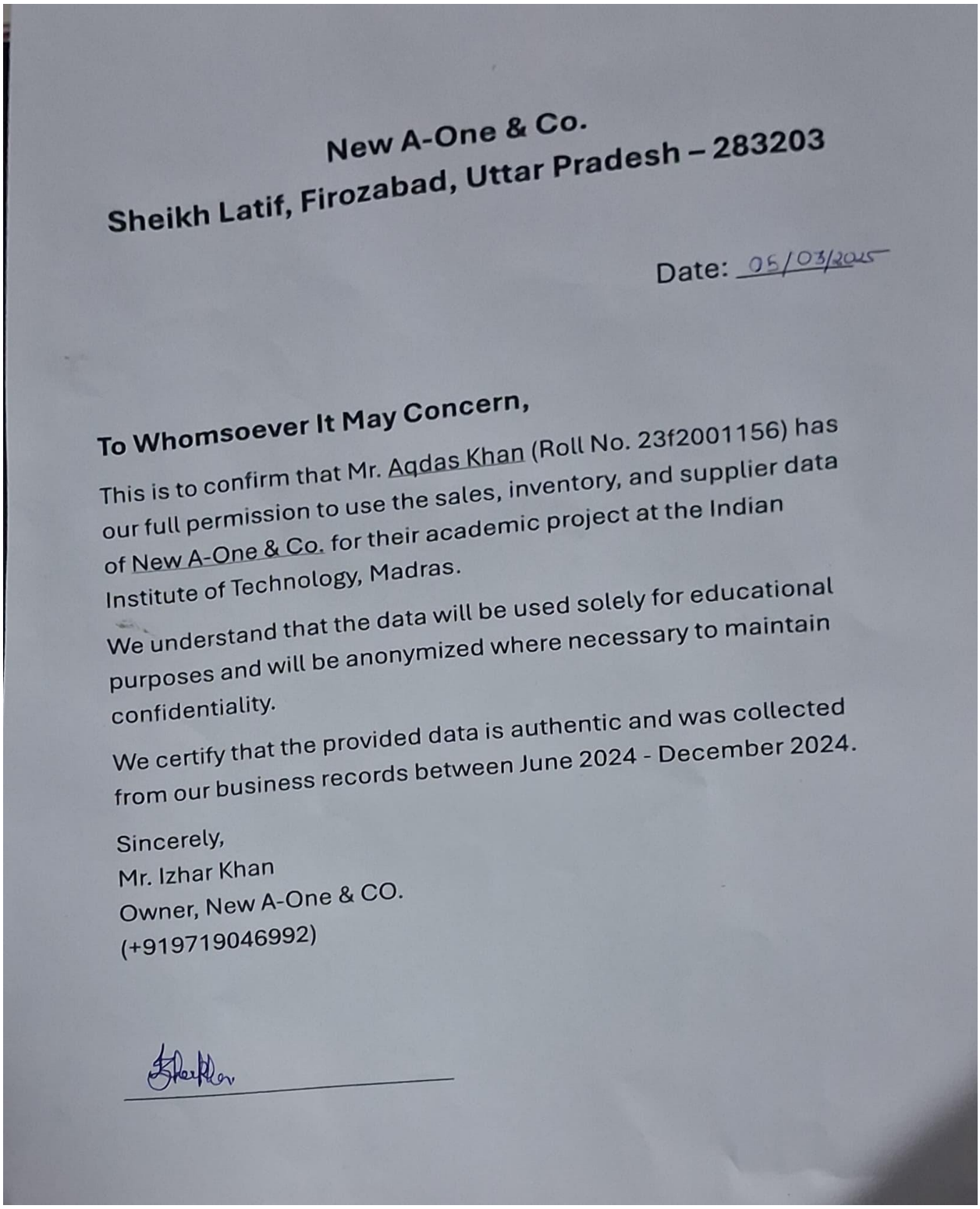
## Methodologies include:

- Descriptive analytics (Excel) to classify inventory turnover and customer payment patterns.
- Predictive modeling (Python/Scikit-Learn) to score customer creditworthiness.
- Visual dashboards (Tableau) to monitor supplier performance metrics.

Expected outcomes include reduction in overstocking costs, improvement in payment cycles, and streamlined supplier procurement. These interventions aim to increase profitability annually while improving operational efficiency.

## 2) Proof Of Originality

i) Letter from Organisation



New A-One & Co.
Sheikh Latif, Firozabad, Uttar Pradesh – 283203

Date: 05/03/2025

**To Whomsoever It May Concern,**

This is to confirm that Mr. Aqdas Khan (Roll No. 23f2001156) has our full permission to use the sales, inventory, and supplier data of New A-One & Co. for their academic project at the Indian Institute of Technology, Madras.

We understand that the data will be used solely for educational purposes and will be anonymized where necessary to maintain confidentiality.

We certify that the provided data is authentic and was collected from our business records between June 2024 - December 2024.

Sincerely,
Mr. Izhar Khan
Owner, New A-One & CO.
(+919719046992)

ii) Images from firm - link






iii) Video interaction : link

# 3) Metadata

Dataset link

**i) Origin of dataset**

During three onsite visits to New-A-One Bangle Store , I worked closely with owner, Mr. Izhar Khan who is also my father to collect and verify all relevant business data. We extracted information from the store's original handwritten notebooks containing daily sales records (45 transactions), supplier purchase orders (40 entries), and customer credit payment histories. To ensure accuracy, we physically verified current inventory levels, product design specifications, and pricing structures against the recorded data. I manually transcribed all this information into customized Excel spreadsheets, which Mr. Izhar Khan personally reviewed for correctness. Throughout the process, we maintained regular discussions to validate findings and explore practical ways to implement data-driven improvements in the store's operations, with Mr. Khan providing valuable insights into the business context behind the numbers.

ii) Metadata

- **Supplier Data Sheet**
  Tracks 40 purchase transactions from June-Dec 2024 across 5 bangle designs showing supplier details (e.g. NavJivan handles 60% DL/ML units) and consistent costs . Includes design-wise totals, with AL being most purchased Some columns remain unused but provide design code explanations

- **Sales Data Sheet**
  Records 45 sales transactions with design quantities and pricing (₹164-176/unit). Features a month-wise breakdown and identifies Shariq Bangle Store as top buyer.

- **Customer Payment History**
  Profiles 9 customers' payments, revealing ₹2.56L outstanding (20% of billings), led by Shariq. Highlights perfect payers (Durga) vs delinquents (Owais: 100%

unpaid). Purchase frequency ranges 1-16 transactions, showing inverse correlation with payment compliance.

- **Monthly Summary**
  Serves as executive dashboard with June-Dec breakdowns. Designed for quick performance snapshots but currently lacks visualizations.

**Metadata Overview**

| Dataset | Records | Time Period | Key Variables | Source |
|---|---|---|---|---|
| **Supplier Data** | 41 | Jun–Dec 2024 | Design, Units, Cost, Supplier, Date | Supplier invoices |
| **Sales Data** | 45 | Jun–Dec 2024 | Design, Customer, Revenue, Quantity Sold | Store invoices |
| **Payment History** | 10 | As of Dec 2024 | Customer, Total Amount, Paid, Pending | Owner's ledger |
| **Formula Driven Monthly Sales aggregation** | 7 | June-Dec 2024 | Months, Units Sold ,Total Revenue Generated | Sales Data |

# 4) Descriptive Statistics

i) **Supplier Data Analysis**

The supplier data reveals significant purchasing patterns across the five bangle designs. A total of 21,101 units were procured from June to December 2024, with AL (AnjuLal) designs dominating at 8,375 units . Purchasing activity peaked in October, total 3635 units acquired, suggesting seasonal demand fluctuations. The average procurement cost remained stable at ₹132 per unit, with Super Glass Works commanding slightly higher prices (₹134/unit) for AL designs. NavJivan Glass Works emerged as the primary supplier, handling almost 60% of DL and ML designs, while maintaining consistent pricing at ₹130/unit across these categories.

Supplier performance metrics highlight operational strengths and weaknesses. NavJivan Glass Works demonstrates remarkable consistency in pricing and supply reliability for DL/ML designs, but potential quality concerns emerge when comparing purchase volumes to sales performance. Super Glass Works, while slightly more expensive, shows strong alignment between AL design purchases and sales performance. The data reveals October as a critical month for inventory planning, with AL design purchases spiking to 2,633 units while maintaining strong sales conversion rates.

ii) **Sales Data Analysis**

Sales performance analysis shows the store sold 18,344 units during the period, with AL designs again leading at 8,068 units .The premium pricing of AL designs (₹176/unit) generated ₹14.2 lakhs in revenue, outperforming DL designs which sold 5,555 units at ₹164/unit. Shariq Bangle Store accounted for 43% of total sales volume through 16 transactions, establishing itself as the store's most valuable customer. Monthly sales remained robust throughout the period, with no single month accounting for less than 12% of total sales, indicating consistent demand.

**iii) Costumer Payment History**

Financial examination of customer payments reveals critical cash flow insights. Total outstanding receivables stand at ₹2.56 lakhs, representing 20% of total billings. Shariq Bangle Store carries the largest overdue balance (₹2.56 lakhs), while Durga Bangle Store demonstrates perfect payment compliance. The data shows an inverse relationship between purchase frequency and payment reliability - frequent buyers like Shariq (16 purchases) maintain only 80.4% payment compliance, compared to occasional buyers like Durga (4 purchases) with 100% compliance. This pattern suggests the need for revised credit policies for high-volume customers.

**iv) Monthly Aggregation Of sales performance**

Sheet 4 summarizes monthly sales performance from June-December 2024, showing December as the peak month (4322 units sold, ₹7.2 lakhs revenue) and June as the weakest (1,444 units, ₹2.4 lakhs). Average monthly revenue was ₹3.56 lakhs with

consistent ₹194/unit pricing across months. October recorded the highest volume (3,315 units) while November showed better revenue efficiency .

Formula-based calculations (SUMIF) dynamically link to 45 sales transactions for real-time updates. The data reveals a 120% sales volume growth from June to December, highlighting strong year-end demand patterns for inventory planning.

# 5) Analysis Process

1. **Data Preparation and Cleaning**

   The analysis began with a thorough review of the dataset's structure and quality. After manually adding records to excel, the Excel files were examined for missing values, inconsistencies, and formatting issues. Key steps included standardizing date formats across sheets, verifying numeric calculations and cross-checking the supplier and customer names for uniformity. Special attention was paid to the "No info" customer entries in sales data and empty columns in supplier records. The month column in Sales data was validated to ensure proper formula functionality, while payment history data was checked for arithmetic accuracy in outstanding balance calculations.

2. **Exploratory Data Analysis (EDA)**

   Initial exploration focused on understanding purchasing and sales patterns through basic statistical measures. For inventory analysis, design-wise purchase totals were compared against sales figures to identify overstocking or understocking situations. The EDA revealed that AL designs accounted for 41.7% of purchases but 44% of sales, suggesting strong demand. Temporal analysis showed purchasing peaks in October and December, while sales remained consistently high throughout the period. Supplier performance metrics were developed by analysing cost consistency and purchase volumes across manufacturers.

3. **Financial Performance Assessment**

   The payment history data was analysed to evaluate the store's cash flow health. Customer payment compliance rates were calculated by comparing amounts paid against total billings, revealing that 20% of receivables remained outstanding. High-value customers like Shariq Bangle Store were identified as both the largest revenue source and the

biggest credit risk. Profit margins were computed by comparing supplier costs (₹130-134) against sales prices (₹164-176), showing AL designs delivered the highest margins. The analysis also highlighted that Durga Bangle Store represented the ideal customer profile with full payment compliance.

4. **Operational Insights Generation**

   Cross-sheet analysis connected supplier data with sales performance to identify supply chain efficiencies and outliners. NavJivan Glass Works' dominance in DL/ML supply was contrasted against their consistent pricing but potential quality issues suggested by sales returns. The month-wise sales breakdown in Sheet4 helped identify seasonal trends, while purchase frequency analysis from the payment sheet informed customer segmentation strategies. Notably, the data showed that frequent buyers tended to have poorer payment compliance, suggesting a need for revised credit terms for high-volume customers.

5. **Visualization and Reporting Strategy**

   For customer payment analysis, a stacked bar chart will compare total billed amounts against paid/outstanding balances per customer, highlighting credit risks . Design performance will use a pie chart to show AL's dominance (44% sales share) and DL/ML's relative contributions. Revenue trends will be tracked via a line chart plotting monthly totals (June–Dec), emphasizing December's peak (₹5.24L) and growth patterns. Pareto analysis of inventory show that AL is most valuable and profitable Design.

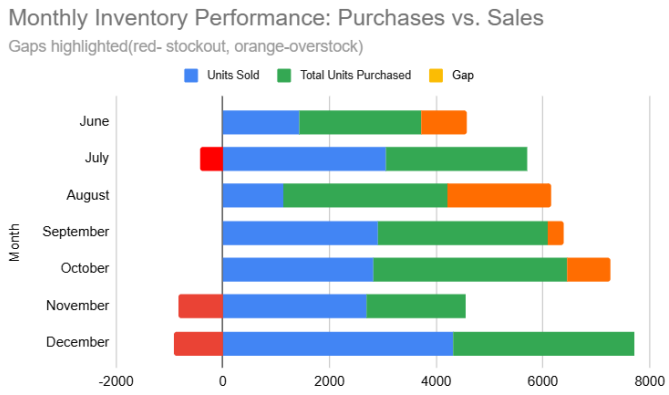
# 6. Results and findings

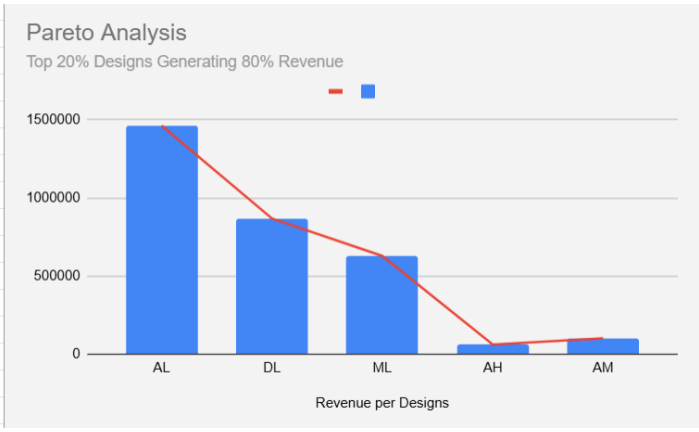
i) Inventory Optimisation Analysis

The analysis revealed a clear revenue concentration in AL (AnjuLal) and DL (DoriaLal) designs, which together accounted for 80% of total revenue despite representing only 70% of purchases. AL designs showed particularly strong performance, generating

Almost 50% of revenue from just 39% of purchased units, indicating superior profitability. However, significant inventory mismatches were identified, with ML designs showing an 8% overstock ,1950 overstocked units in August and stockout in the month of November . These findings demonstrate the need for more precise

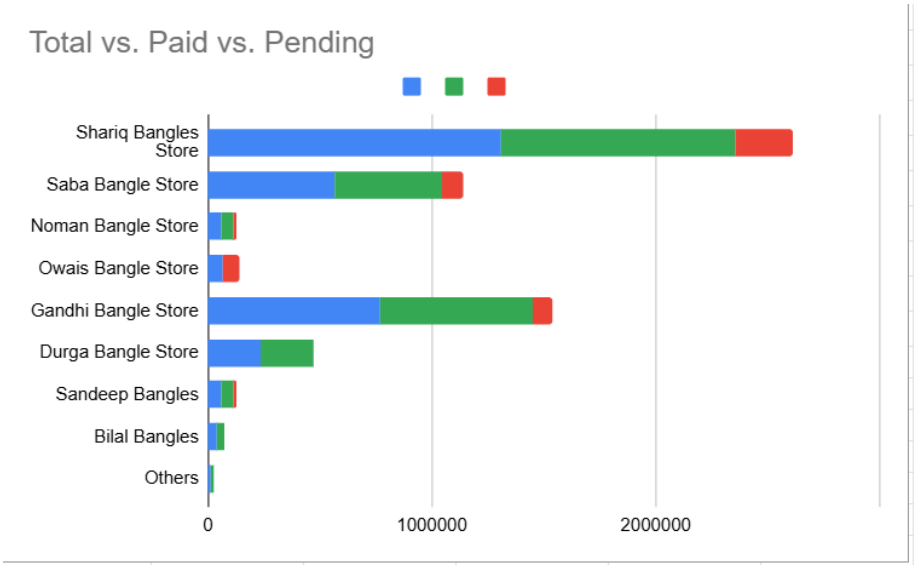
demand forecasting and inventory rebalancing, particularly to capitalize on AL's strong market performance while reducing slow-moving ML stock



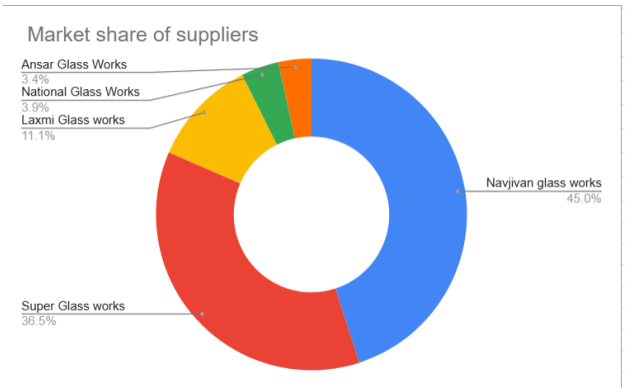

ii) Credit Risk Assessment Finding

Payment behavior analysis uncovered concerning trends among high-volume customers. Shariq Bangle Store, despite being the largest customer with 16 transactions, accounted for ₹2.56 lakh in overdue payments ,20% of total receivables. Owais Bangle Store had 100% of its ₹68,992 outstanding. The data revealed an inverse relationship between purchase frequency and payment reliability - while Shariq made frequent purchases, it maintained only 80.4% payment compliance, compared to Durga Bangle Store's perfect 100% payment record across its 4 transactions. This pattern suggests the current credit system disproportionately favors high-volume but higher-risk customers, necessitating revised credit terms to improve cash flow stability.

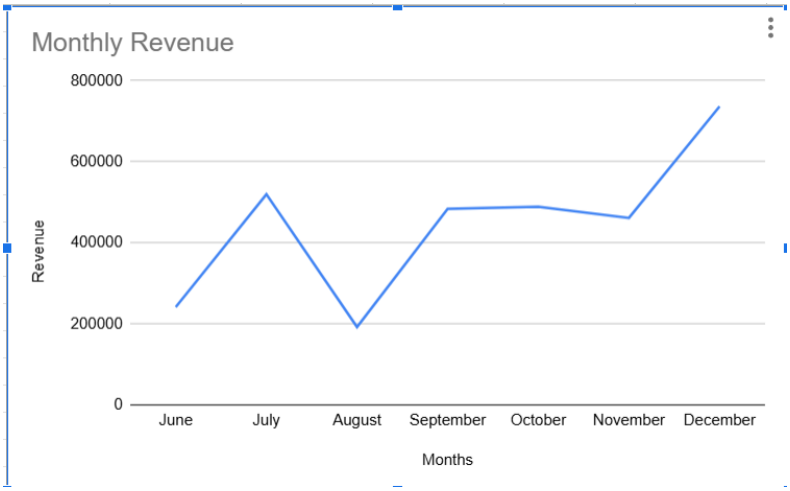



iii) Supply Chain Efficiency

Supplier evaluation showed NavJivan Glass Works dominates procurement, supplying 60% of DL/ML units at ₹130/unit, but with noticeable quality inconsistencies. Super Glass Works emerged as a premium but reliable supplier for high-margin AL designs ,₹134/unit, best order accuracy, while Laxmi Glass Works showed concerning longer lead times. This supplier concentration risk, particularly for DL/ML designs, combined with the quality and delay issues, underscores the need for supplier diversification and performance-based contracting to mitigate operational vulnerabilities in the supply chain.




iv) Monthly Sales trends and revenue analysis




The analysis reveals strong seasonal trends, with revenue peaking in December (₹7.3L) and October (₹4.89L) due to festive and wedding demand, driven predominantly by AL designs . June was the weakest month , reflecting off-season sluggishness, while November showed exceptional revenue efficiency  likely due to bulk orders. A 120% growth from June to December underscores escalating year-end demand, though inventory mismatches emerged as Decembers's stockouts (sales > purchases) and August's overstocking (purchases > sales) highlight the need for demand forecasting. The consistent ₹194/unit revenue confirms stable pricing, but AL's premium margins ₹176/unit prove critical for profitability. Strategic takeaways include scaling AL/DL stock before peak months and promoting slow-moving designs ML/AH during lulls to optimize cash flow.

This analysis highlights critical trends in inventory, credit risk, and supply chain performance, revealing actionable opportunities to optimize operations.  The data underscores the dominance of AL/DL designs in revenue generation, persistent payment delays from high-volume buyers, and supplier inefficiencies requiring strategic adjustments. These findings set the foundation for implementing targeted solutions to enhance profitability and operational stability in the final project phase.